

Total No. of Questions : 10]

SEAT No. :

PA-342

[Total No. of Pages : 3

[5927] : 226

B.E. (Mechanical/Mechanical Sandwich)

MECHANICAL SYSTEM DESIGN

(2015 Pattern) (Semester - II) (402048)

Time : 3 Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve question Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of logarithmic tables slide rule, Mollier charts, electronic pocket calculator and steam tables is allowed.
- 5) Assume suitable data, if necessary.

Q1) a) Draw structural diagram for following structural formulae. **[6]**

i) 3 (1) 3 (3)

ii) 2 (1) 3 (2) 2 (6)

- b) Find the reliability of operating period of 1000 Hrs for a device having failure rate of 1×10^{-6} failure/hrs .If there are 1000 items in the test, how many failures are expected in 1000 hrs. **[4]**

OR

Q2) a) A 2×2 drive is required for transmitting speed starting from 600 rpm with. Geometric progression ratio 1.4. draw suitable structural diagrams. **[6]**

- b) Explain the difference between Design and Natural Tolerance. How the designer would select the tolerances for minimum rejection of the Components. **[4]**

Q3) A belt conveyor is to be designed to carry the bulk material at the rate of 300×10^3 kg/hr with following details

Density of bulk material = 800 kg/ m³

Angle of repose of bulk material = 15 degree

Belt speed= 10 Km/hr

Material factor for plies, K1 = 2.0

Belt tension and arc of contact factor, k2 = 63

No of plies for belt = 4

Determine:

- a) The suitable width of belt
- b) The diameter and length of drive pulley

OR

P.T.O.

- Q4) a)** State the advantages, disadvantages and applications of conveyor as material handling equipment. [4]
- b)** Horizontal belt conveyor is to be used transporting material with mass density of 3 tons/m³. the belt is 1m wide and has speed of 2 m/s determine the capacity of conveyor if surcharge angle is 20 degrees. [6]

- Q5) a)** Derive the expression to determine the principal stresses at the inner surface of thick cylinder. [8]
- b)** A tube of inner and outer diameter 50 mm and 75 mm respectively is reinforced by shrink fitting another tube of 100 mm diameter. This assembly is to withstand an internal pressure of 35 MPa. If the shrinkage is such that the maximum tangential stress in both the tubes is same, [10]
Calculate:
i) The shrinkage pressure
ii) The original dimensions of the tube.
Assume $E = 207 \text{ kN/mm}^2$.

OR

- Q6) a)** What is the need of autofrettage? Explain any method of autofrettage in detail along with sketch. [6]
- b)** A cylindrical pressure vessel with 1500 mm inside diameter and 150 mm thickness is subjected to internal pressure of 1.5 N/mm². A nozzle of 300 mm inside diameter and 12 mm thickness is to be provided in the shell. Welds joints efficiency for shell as well as nozzle is 85 % , while corrosion allowance is 2 mm. The nozzle projects 20 mm outside and 30 mm inside the shell. If the allowable tensile stress for the shell and nozzle material is 120 N/mm², determine whether the reinforcing is adequate, if not find the dimensions of reinforcing pad, if the plate thickness is 150 mm. [12]
- Q7) a)** What is the cross section of connecting rod for low speed and high speed engines? Explain with sketch. [6]
- b)** Following data is given for piston of four stroke diesel engine: [10]
i) Cylinder bore = 250 mm;
ii) Maximum gas pressure = 4 N/mm²
iii) Bearing pressure = 15 N/mm²
iv) Length of piston pin in bush of small end = 0.45 D;
v) Ratio of inner to outer diameter of piston pin = 0.6;
vi) Mean diameter of piston boss = 1.4 × Outer diameter of piston pin;
vii) Allowable bending stress of piston pin = 84 N/mm²

Calculate:

- 1) The outer diameter of piston pin;
- 2) The inner diameter of piston pin;
- 3) The mean diameter of piston boss
- 4) Check design for bending stress

OR

Q8) Design a cylinder and cylinder head for a four stroke C.I. engine with the following data: **[16]**

- Brake Power = 5 kW;
- Engine speed = 1200 rpm; Indicated mean effective pressure = 0.3 N/mm²;
- Maximum gas pressure = 3.5 N/mm²;
- Mechanical efficiency = 80%;
- Compression ratio = 12;
- Boring factor, $C_1 = 4$ mm;
- Cylinder head thickness constant $K_1 = 0.35$ and Allowable tensile stresses are.

Name of Part	Cylinder Liner	Cylinder Head	Studs
Material	Alloy Cast Iron	Alloy Cast Iron	Alloy steel 40Ni3
Allowable tensile stress N/mm ²	40	40	70

Q9) a) Explain methodologies of optimum design in detail with suitable example. **[6]**
b) A tensile bar of length 400 mm is subjected to constant tensile force of 4000 N. If the factor of safety is 2 design the bar with the objective of minimizing the material cost out of the following materials. **[10]**

Material	Mass density, ρ (Kg/m ³)	Material Cost, C (Rs/N)	Yield Strength S_{yt} (MPa)
Plain carbon steel	7800	28	400
Alloy steel	7850	150	900
Al alloy	2800	140	150
Titanium alloy	4500	2200	800

OR

Q10) In a light weight equipment a shaft is required to transmit 40 KW power at 425 rpm. The required stiffness of the shaft is 90 N-M/ Degree. The factor of safety based on yield strength in shear is 1.5. Using the maximum shear stress theory design the shaft with the objective of minimizing the weight out of the following materials. **[16]**

Material	Mass density, ρ (Kg/m ³)	Material Cost, C (Rs/N)	Tensile Yield Strength s_{yt} (MPa)	Modulus of rigidity, G N/mm ²
Alloy steel	7800	7.5	450	82×10^3
Al alloy	2800	9	150	27×10^3
Titanium alloy	4500	150	800	41×10^3
Magnesium alloy	1800	10	100	17×10^3

